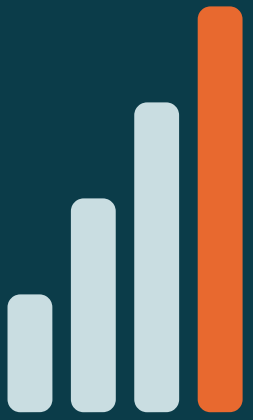
Keeping the Customers Worth Keeping



A data-driven retention proposal for Company A · Proof of Concept

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GCI World 2026 · Final Assignment

Executive summary

Broad retention offers lose money — a targeted, value-based programme makes it profitable

1

The problem

Customers leave in a saturated market. Company A cannot tell who will leave, so retention money is spent blindly.

2

What the data shows

Churn rises sharply with ageing handsets (>1 year), falling usage, and the 11–12 month contract end.

3

What we built

A churn-risk score (ensemble of 3 tree models). It ranks customers correctly 71% of the time (ROC-AUC 0.71, up from 0.70).

4

What we propose

Each month, contact only customers whose expected saved profit beats the offer cost, with an offer that fits their reason to leave.

82.2% vs 16.9%

churn in the model's top-10% vs bottom-10% risk group

−$4.09M

per year if we give everyone an offer

+$0.13M / yr

with our targeted rule (ROI 0.64), base case

Source: Company A data (100,000 customers), own analysis. Profit on a 100k-customer base at a realistic 2% monthly churn; assumptions on slide 13.

Why churn matters in mobile telecom

Growth now means keeping customers, not finding new ones

5.6 bn

unique mobile subscribers worldwide (end-2023) — about 69% of people. The market is saturated. [3]

5–25×

more expensive to win a new customer than to keep an existing one. [2]

+25–95%

profit increase linked to a 5-point rise in customer retention. [1]

What this means for Company A

- Every lost customer must be replaced by taking one from a competitor — at a much higher cost.
- Handset upgrades and new technology (5G) are moments when customers compare operators [4].
- Research shows churn can be predicted from usage, billing and handset data — and that the value comes from the campaign it drives, not the model itself [5][6].

Sources: [1] Reichheld & Schefter 2000; [2] Gallo 2014 (HBR); [3] GSMA Mobile Economy 2024; [4] Ericsson Mobility Report; [5] Verbeke et al. 2012; [6] Neslin et al. 2006.

The data at a glance

100,000 customers, 100 facts each — rich, but with gaps and one trap

100,000

customers

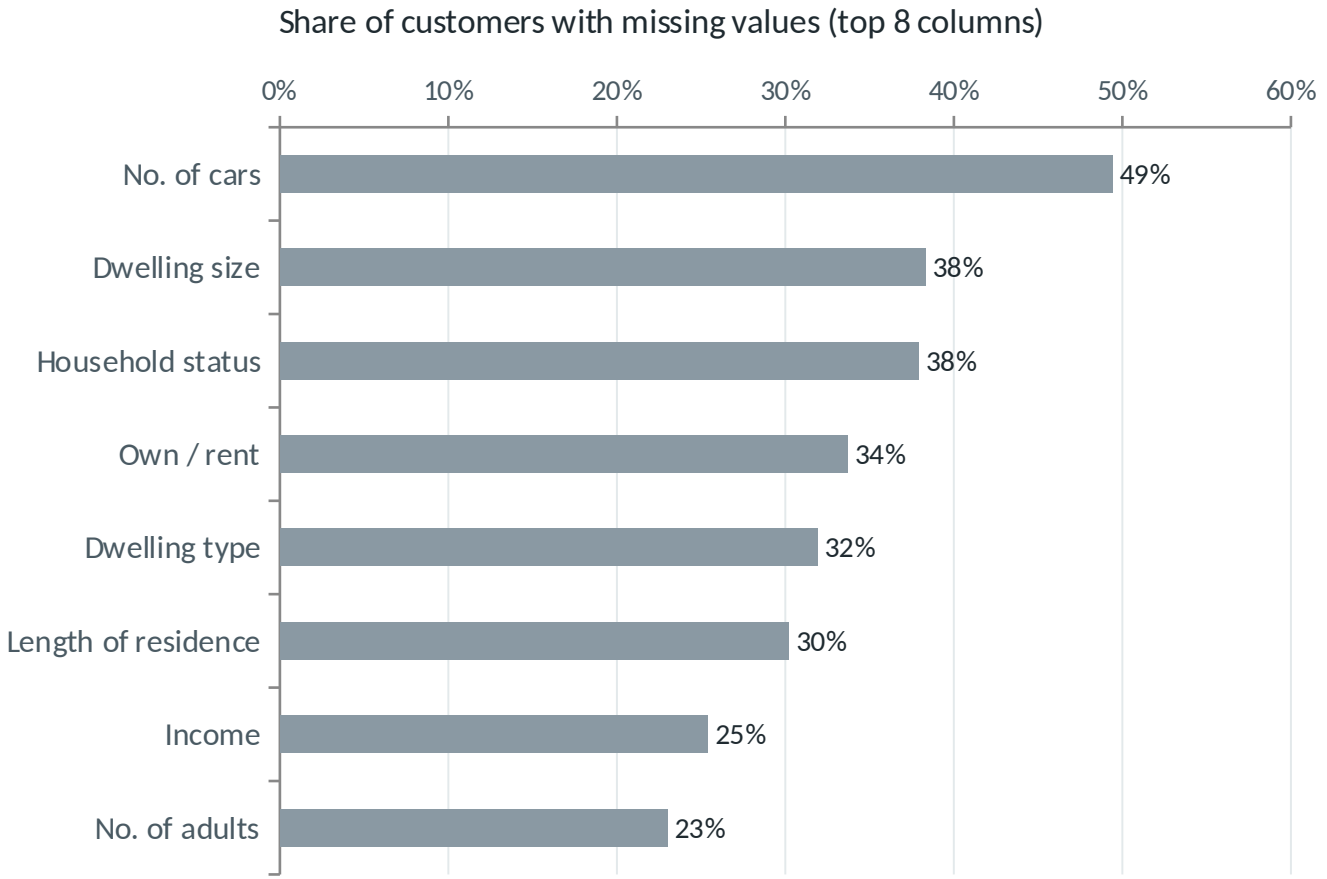
100

columns: usage, billing, handset, household

2 tables

joined 1-to-1 on Customer ID

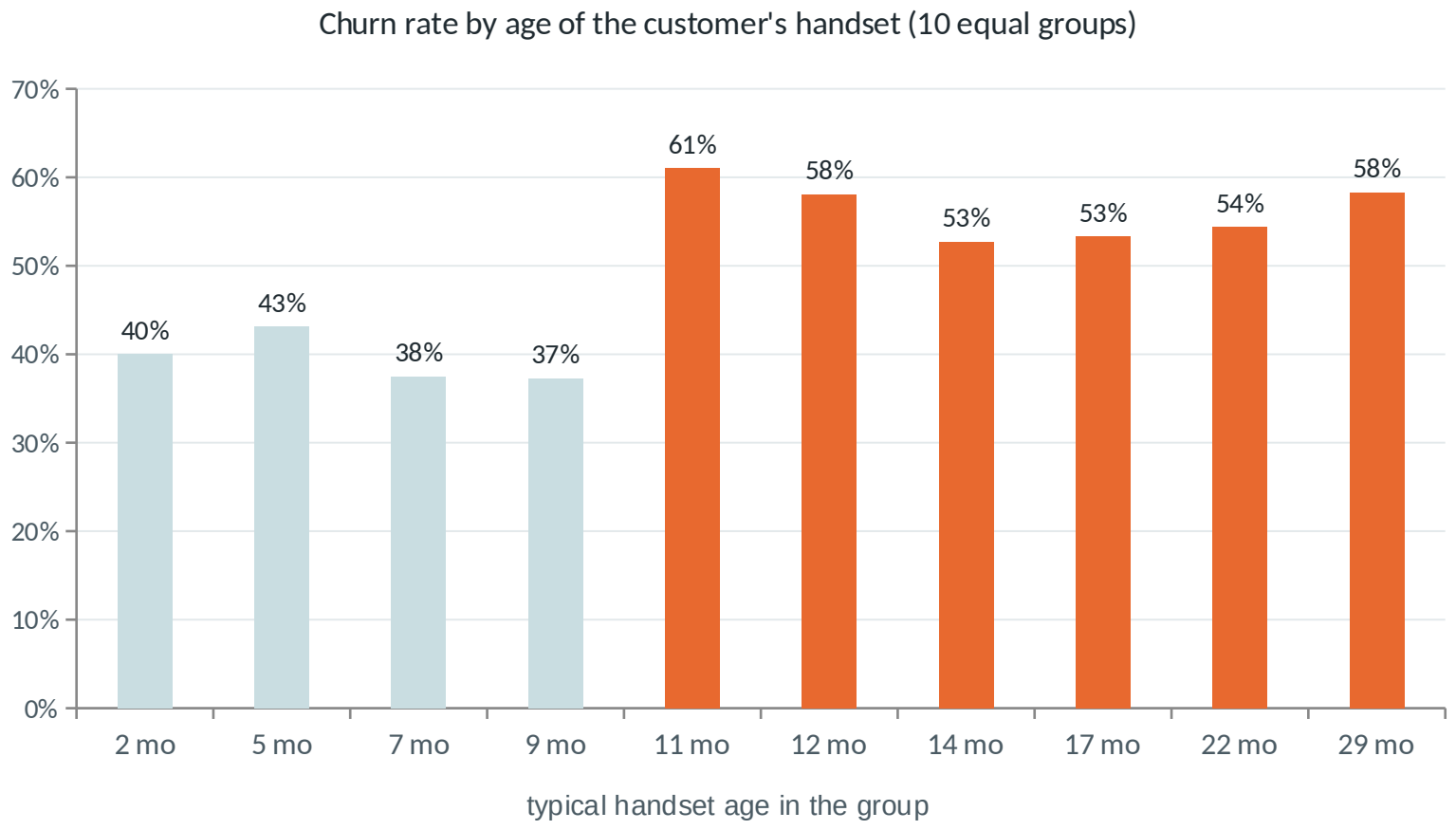
Watch out: 50% of these customers churned, but a real operator loses ~2% a month. The sample was balanced on purpose, so we correct for it in the business case.



Source: Company A data. Missing values are mainly third-party household data; usage and billing data are almost complete. Gaps were kept, not deleted.

Old handsets are the clearest warning sign

After about one year, churn jumps from ~40% to ~55–60%



What it means

- Customers with a handset older than ~11 months are about 1.5× more likely to leave.
- That is when rivals' new-phone deals look most attractive.
- Cheaper and refurbished handsets also churn more (57% vs 36% for the cheapest vs most expensive).
- **Lever: a timely upgrade offer.**

Source: Company A data, own analysis. Each bar = 10% of customers, ordered by handset age.

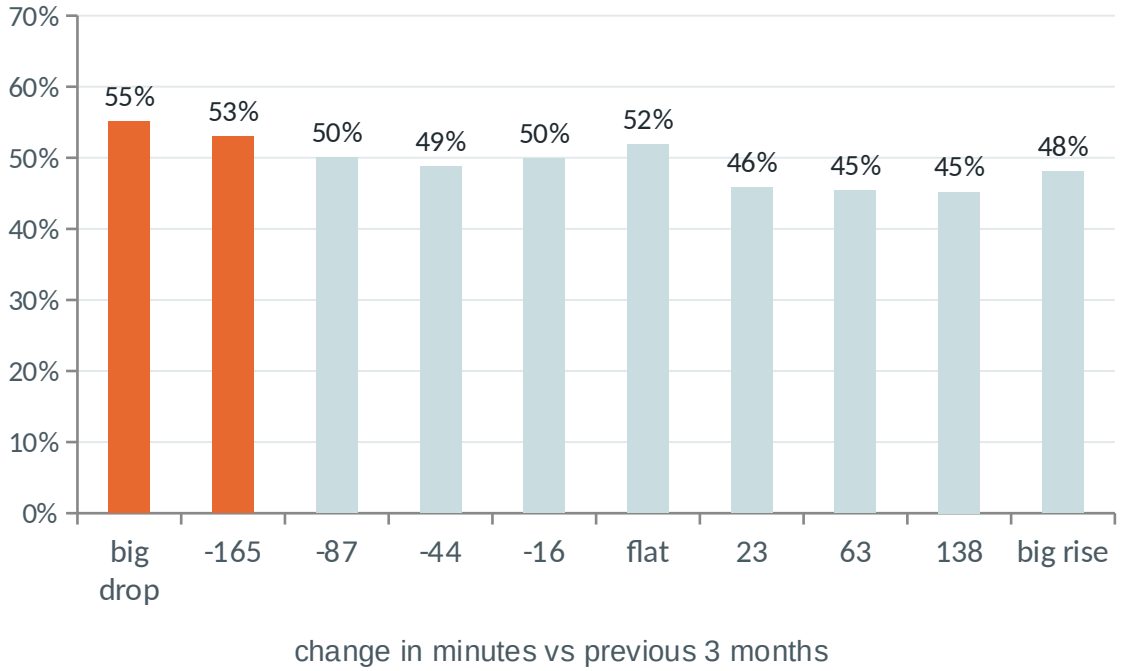
Contract end and falling usage signal departure

Risk peaks at month 11–12, and customers use us less before they leave

Churn rate by months as a customer



Churn rate by change in monthly minutes (10 groups)



Month 11: 65% churn — right as one-year contracts end. Act before month 10.

Usage drop = early warning: 55% churn vs 45% when usage grows.

Source: Company A data, own analysis.

From findings to a clear question

Four ideas tested — three confirmed and turned into one prediction task

What we expected → what the data said

Old handsets drive churn	**Confirmed**	~40% → ~58% after 1 year
Falling usage is an early warning	**Confirmed**	55% vs 45% churn
Contract end concentrates risk	**Confirmed**	65% at month 11
Poor network quality drives churn	**Not supported**	Dropped calls barely matter

The question for the model

Target: will this customer leave in the next 31–60 days? (yes / no)

Why this target: it is linked to levers we can pull, and leaves one month to act.

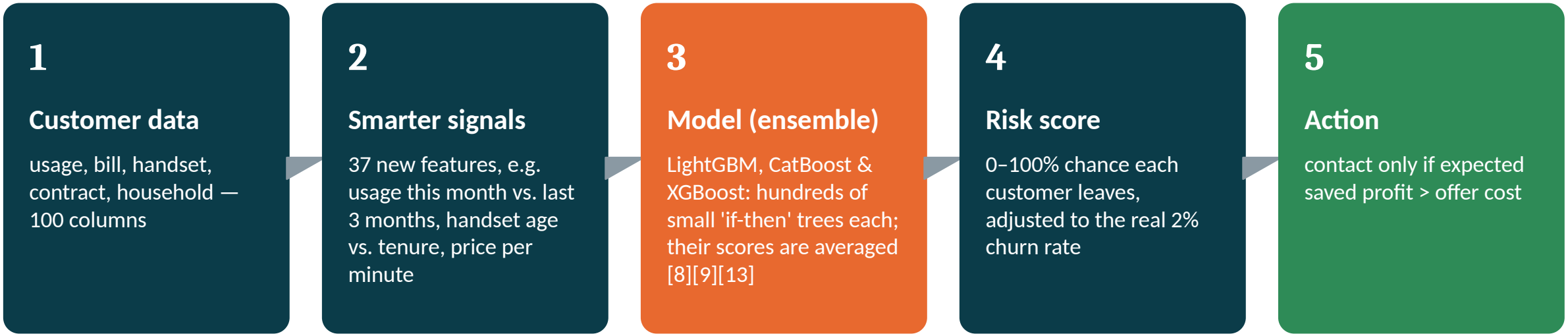
How we score the model: ROC-AUC — how often it ranks a real leaver above a stayer (0.5 = coin toss, 1.0 = perfect). Plus accuracy.

How we score the business: net profit of the retention campaign [5].

Source: own analysis. [5] Verbeke et al. 2012 — profit-driven evaluation of churn models.

How the churn-risk score works

Three models of many small decision rules turn 135 facts into one risk score per customer



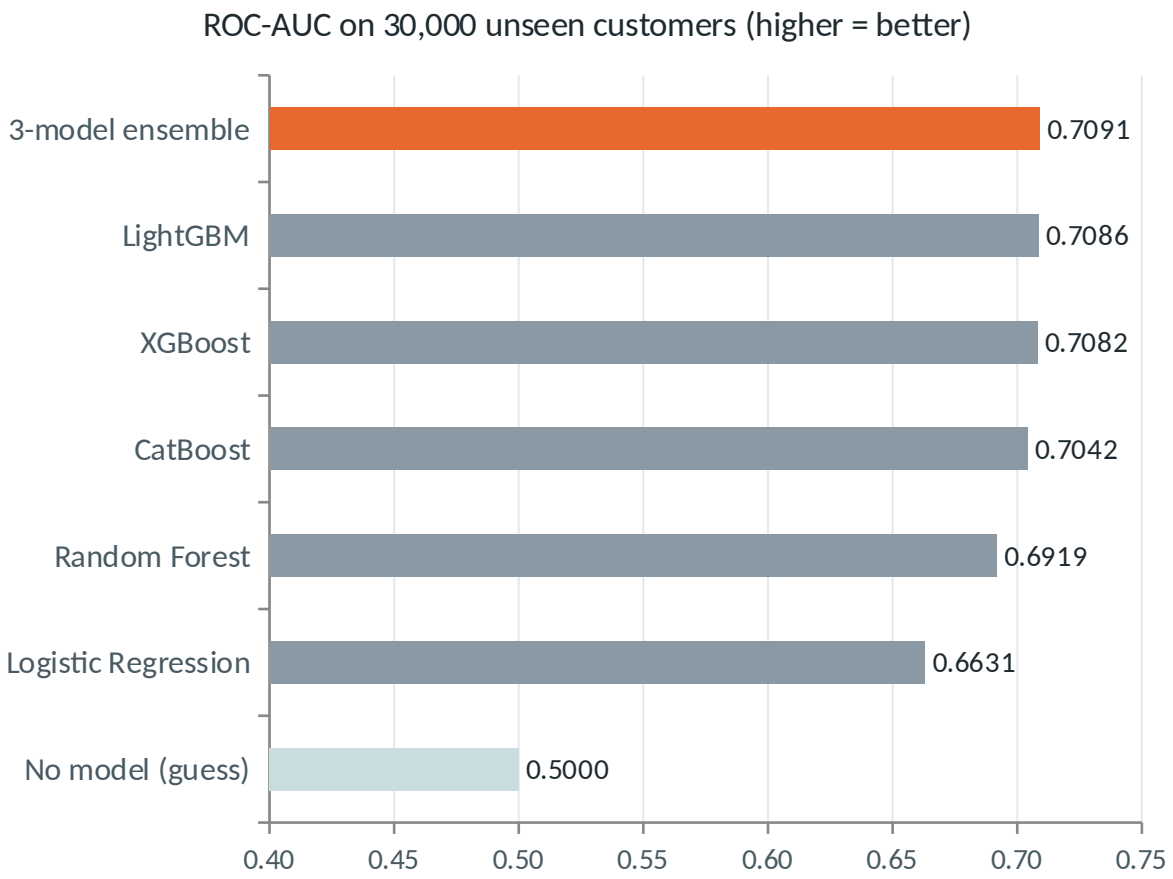
Why this approach

- Tree-based models handle mixed data, missing values and extreme values without manual fixes — exactly what this data has.
- Tested fairly: settings tuned automatically on 70% of customers with 5-fold cross-validation [14], then scored once on the other 30% it never saw.

Methods: [7] scikit-learn; [8] XGBoost; [9] LightGBM; [10] Random Forest; [13] CatBoost; [14] Optuna tuning; [15] ensemble averaging.

Combining three models gives the most accurate score

Best: 3-model ensemble — ROC-AUC 0.709, accuracy 64.8% (first version: 0.703, 64.1%)



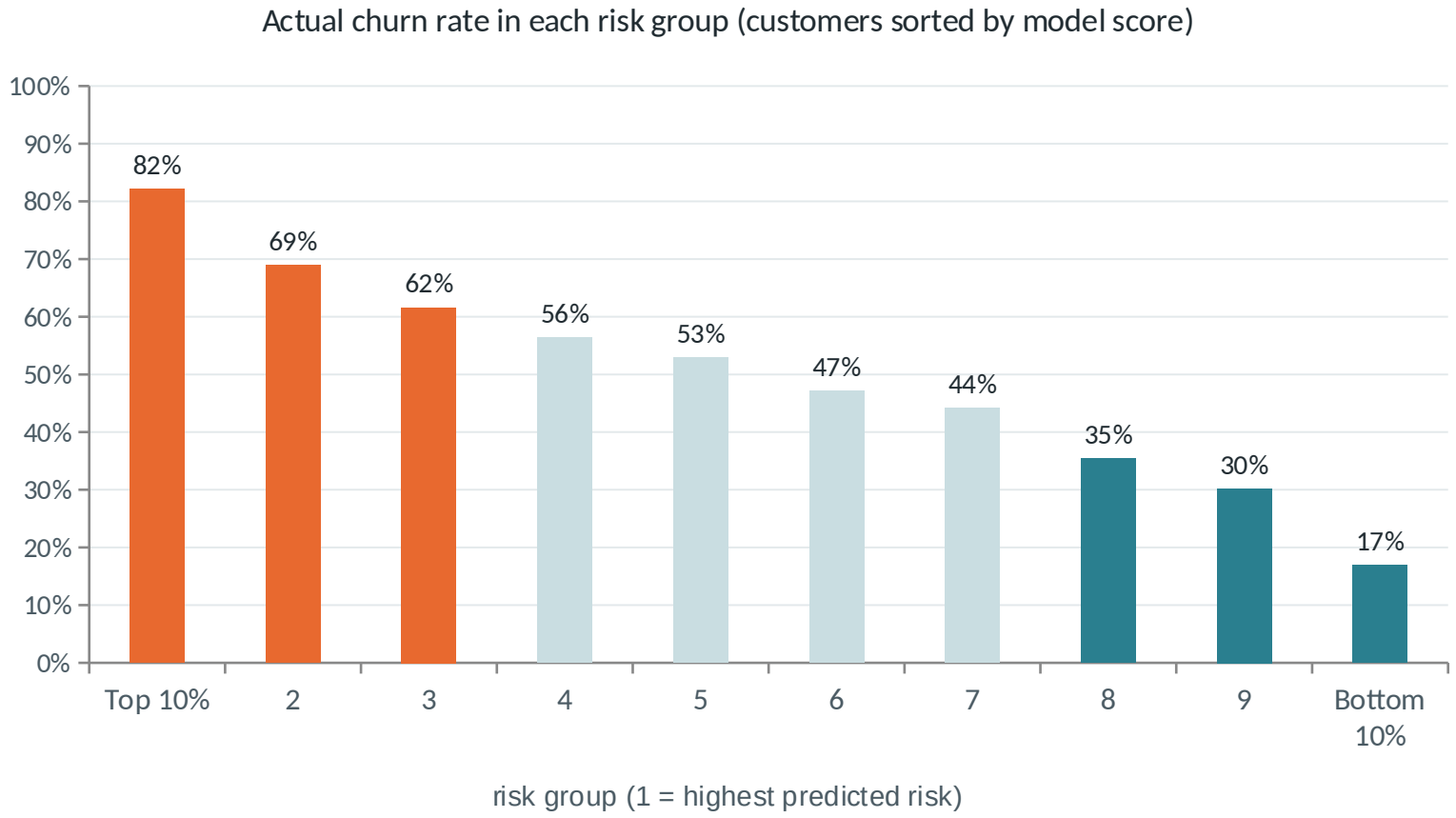
Model	ROC-AUC	Accuracy	F1
3-model ensemble	**0.7091**	**64.8%**	**0.644**
LightGBM	0.7086	64.8%	0.645
XGBoost	0.7082	64.8%	0.644
CatBoost	0.7042	64.5%	0.641
Random Forest	0.6919	63.4%	0.635
Logistic Regression	0.6631	61.6%	0.610
First version (LightGBM)	*0.7025*	*64.1%*	*0.641*
No model (guess)	0.5000	50.4%	0.000

In plain words: pick one customer who left and one who stayed — the model gives the leaver the higher risk score 71% of the time. New usage signals, automatic tuning and combining three models lifted every score vs. the first version. Cross-validation agrees (0.708 ± 0.002): stable, not luck.

Source: own analysis; 70/30 train/test split, 5-fold cross-validation, settings tuned with Optuna [14]. AUC ≈ 0.70 is typical for wireless churn data [6] — some churn is driven by events not in the data.

The score finds the customers who will leave

The 10% highest-risk customers churn 4.9× more often than the 10% lowest-risk



82.2%

of the top-10% risk group actually left

43%

of all leavers are in the top 30% — vs 30% if we picked at random

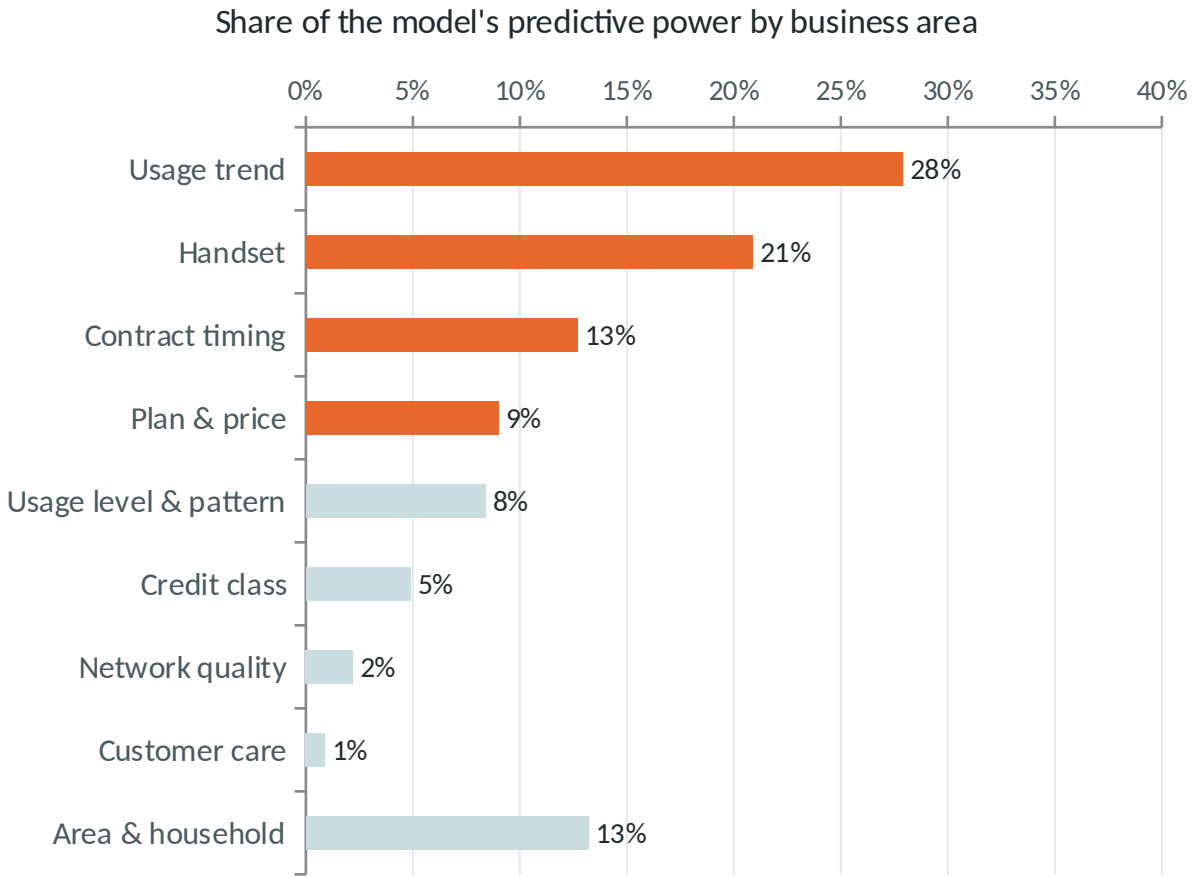
16.9%

churn in the lowest-risk group — no need to spend here

Source: own analysis on 30,000 held-out customers (sample is 50% churners by design).

Why customers leave — and what we can do about it

About 70% of the model's knowledge comes from things Company A can act on



Signal → action

Usage trend: Customer uses us less than usual
→ **Proactive call: right-size the plan**

Handset: Handset older than ~1 year
→ **Upgrade offer tied to renewal**

Contract timing: Month 10–13 of tenure
→ **Early-renewal bonus**

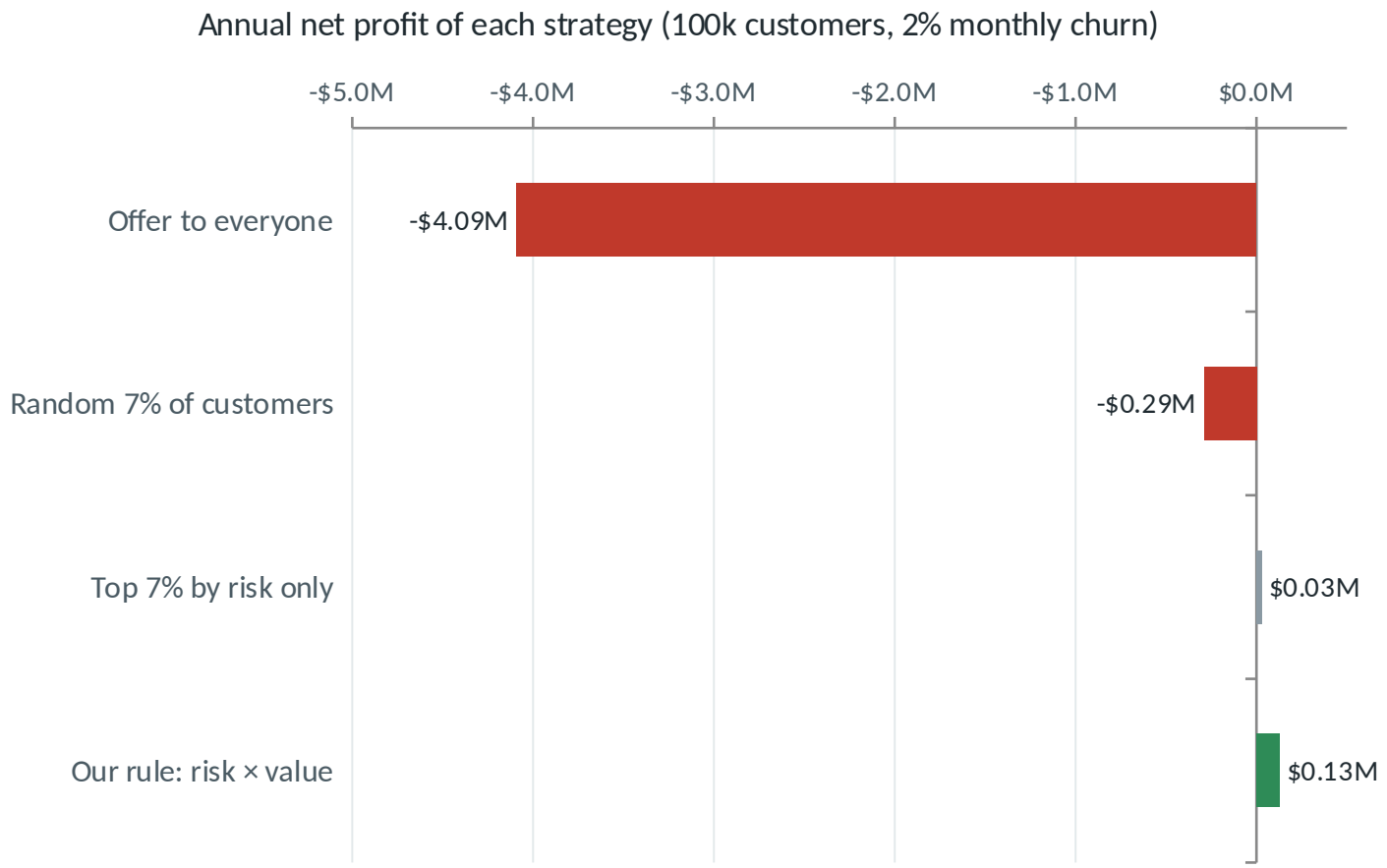
Plan & price: Low monthly plan, cheap handset
→ **Loyalty bundle**

Network quality and customer-care calls add almost nothing — budget should go to handset and plan offers.

Source: own analysis — permutation importance [10] on unseen customers (drop in accuracy when a fact is scrambled). Demographics are not used for targeting decisions.

Who we target decides whether retention pays

Offers to everyone lose millions — targeting risk × customer value turns a profit



Why blanket offers fail

- Only ~2 in 100 customers leave in a month — 98 would have stayed anyway, and many still take the discount.
- The riskiest customers are often low spenders (avg $48/month) — not worth a $25 offer.
- **Our rule picks customers who are both at risk and valuable (avg $125/month).**

Source: own analysis on held-out customers, re-weighted from the 50/50 sample to a 2% monthly churn rate; assumptions on slide 13.

Proposal: a monthly, targeted retention programme

About 3,600 customers a month, each with an offer that fits their reason to leave

Who gets which offer (share of selected customers)

43% — **Upgrade-due:** Handset > 1 year
→ **Discounted upgrade + 12-month renewal**

26% — **Contract-end:** Month 10–13
→ **Early-renewal bonus**

19% — **Other:** High value, other risk
→ **Loyalty credit**

12% — **Usage-decline:** Usage < 70% of normal
→ **Plan right-sizing call**

Expected impact per year (base case)

~625 customers kept

$0.85M revenue protected

+$0.13M net profit (ROI 0.64)

Assumptions: 2% monthly churn · 25% of contacted leavers are saved · 15% of stayers also redeem · $25 offer, paid only on renewal · $1 contact (SMS/app) · 40% margin · saved customers stay 12 months

Source: own analysis; impact scaled to 100,000 customers. Assumptions are estimates to be confirmed in the pilot (slide 14).

The result holds up, and grows with a better save rate

Next step: a 3-month pilot to measure what an offer really achieves

Annual net profit under different assumptions ($M, 100k customers)

Saved ↓ · churn →	1.5%/mo	2.0%/mo	3.0%/mo	5.0%/mo
10% saved	$0.00M	$0.01M	$0.04M	$0.14M
20% saved	$0.04M	$0.07M	$0.22M	$0.65M
25% saved	$0.07M	**$0.13M**	$0.35M	$1.04M
30% saved	$0.10M	$0.21M	$0.51M	$1.48M
40% saved	$0.21M	$0.40M	$0.94M	$2.52M

Bold = base case. The rule never loses much: in weak scenarios it simply contacts fewer customers. In more competitive, prepaid-heavy markets (3–5% monthly churn) the value rises to $0.35–1.04M a year.

Roadmap

1. **Pilot (months 1–3)**
   Split high-value at-risk customers into offer vs. no-offer groups to measure the true save rate.

2. **Scale (months 4–6)**
   Monthly scoring feeds the CRM and call centre; retrain and monitor every month.

3. **Improve (6+ months)**
   Target customers most likely to respond (uplift model); add complaint and competitor data.

Risks: assumptions not yet measured; the model shows links, not proof of cause; behaviour changes over time. Demographics are excluded from targeting.

Source: own analysis (sensitivity of the expected-value rule; $25 offer). Pilot design follows the profit-driven evaluation of churn models [5].

References

All external sources, methods and tools used in this deck and notebook

[1] Reichheld, F. F., & Schefter, P. (2000). E-Loyalty: Your Secret Weapon on the Web. Harvard Business Review, 78(4). hbr.org/2000/07/e-loyalty-your-secret-weapon-on-the-web

[2] Gallo, A. (2014). The Value of Keeping the Right Customers. Harvard Business Review. hbr.org/2014/10/the-value-of-keeping-the-right-customers

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[4] Ericsson (2024). Ericsson Mobility Report. ericsson.com/en/reports-and-papers/mobility-report

[5] Verbeke, W., et al. (2012). New insights into churn prediction in the telecommunication sector: a profit driven data mining approach. European Journal of Operational Research, 218(1), 211–229.

[6] Neslin, S. A., et al. (2006). Defection detection: measuring and understanding the predictive accuracy of customer churn models. Journal of Marketing Research, 43(2), 204–211.

[7] Pedregosa, F., et al. (2011). Scikit-learn: Machine Learning in Python. JMLR, 12, 2825–2830.

[8] Chen, T., & Guestrin, C. (2016). XGBoost: A Scalable Tree Boosting System. Proc. KDD 2016, 785–794.

[9] Ke, G., et al. (2017). LightGBM: A Highly Efficient Gradient Boosting Decision Tree. NeurIPS 30.

[10] Breiman, L. (2001). Random Forests. Machine Learning, 45(1), 5–32.

[11] GCI World 2026 course materials: Final Assignment tutorial notebook and Company A Dataset Overview.

[12] Generative-AI assistance (Claude Code), used for analysis support, code, slide drafting and the model-accuracy improvements: claude.ai/code/session_01NK6VUjaBPQ9mz2bW8AwXsP; claude.ai/code/session_01NK6VUnLRE5S8Gnv2BQTJqH

[13] Prokhorenkova, L., et al. (2018). CatBoost: unbiased boosting with categorical features. NeurIPS 31.

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[15] Dietterich, T. G. (2000). Ensemble Methods in Machine Learning. Multiple Classifier Systems, LNCS 1857, 1–15.

Data: Company A telecom dataset (Client.csv, Record.csv) provided by GCI World 2026. Full reproducible analysis: Shafayet_Jamil.ipynb.